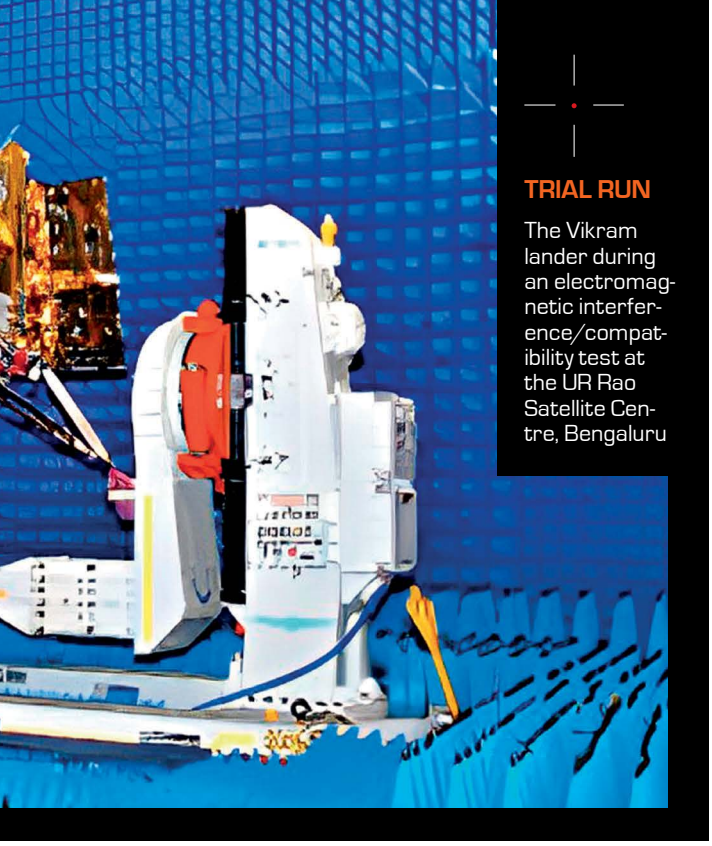


TRIAL RUN

The Vikram lander during an electromagnetic interference/compatibility test at the UR Rao Satellite Centre, Bengaluru

Covid played spoiler for almost a year, seriously disrupting the work on correcting Chandrayaan-2's errors. Top-level changes also began happening. For one, Muthusamy Sankaran took over as director of URSC—the nodal centre that is responsible for the integration of the spacecraft, including for Chandrayaan-3—in June 2021. The clear-thinking Sankaran, 59, who had worked both on the Chandrayaan and the Mangalyaan (Mars) projects, concluded that the chief learning from the failed effort was that the lander was more than just a spacecraft. It was also a satellite with a reusable rocket and an aircraft with a sense of awareness of how and where to land. "It's really a three-in-one craft," explains Sankaran, "and once we treated it in this holistic way, we began to have a better understanding of how to make it work successfully." He also realised that the telemetry data from the crashed lander was limited and would not provide any major clue to what went wrong. He, therefore, pushed for recreating what happened in the final minutes of the craft's descent and then came up with robust solutions to correct it.

Somanath succeeded Sivan as ISRO chairman in January 2022. In addition to a cool-headed approach, Somanath combined his dynamism with a derring-do attitude that enthused the members of his team. He also brought to the table the vast experience of designing and handling complex missions, including the previous Chandrayaan ones. Somanath introduced two major changes in the management style of the project. The first, he termed a failure-oriented approach, a variation of the old Murphy's Law according to which everything that can go wrong will go wrong. He clarified that such an approach did not come from a place of pessimism, it was rather a departure from the assumption of success and hope of all going well, and instead mapping all that could go wrong to then work out how to eliminate glitches or introduce backup systems.

The second was, as he told his team, "We will not go for the launch of Chandrayaan-3 unless there is 100 per cent satisfaction of all the teams involved." That mandate saw every aspect of the lander and rover being stripped threadbare all over again and tested. From descent to touchdown, there are 170 such critical parameters, each of which was tested and retested till each team involved built enormous confidence levels. Veeramuthuvel recounts with a smile that when he reported to Somanath he was confident that the problem had been sorted, he told him, "It is not for you to say that you are confident. The team handling it should be saying it." Somanath also tapped into the vast pool of retired ISRO scientists to review the process and guide the teams. Prominent among them was former ISRO chairman A.S. Kiran Kumar, who provided invaluable insights into the modifications needed for success.

TAKE-NO-CHANCES APPROACH

Each of the key systems teams then got down to identifying and correcting the errors and in the process introduced more than 20 major changes across the board after the Chandrayaan-2 experience. In the propulsion systems, for instance, it examined in depth what could have caused the four engines to overperform in the rough braking phase. To do so, LPSC's Narayanan had his team strap the four engines on a platform at his centre at the Valiamala complex in Thiruvananthapuram and fire them for the entire duration of Vikram's planned descent. It was here that the team detected a missing electronic pulse that controls the valves regulating the flow of propellants. It had resulted in a small excess that caused the engines to overperform. This was corrected by ensuring that the fuel flow was a little more sluggish to compensate for any errors. Then, the four engines were strapped on to a Vikram platform that had all the vital systems such as sensors and control systems and hung on a 60-metre-tall tower at the Sriharikota launchpad. These were fired to see if the vibrations caused by the rocket engines were disturbing the sensors and to prevent the sloshing of fuel impacting the spacecraft's centre of gravity while in flight. Those issues were sorted by bolstering them with dampeners.

When it came to the guidance and control systems, which constituted the brains of Vikram, since the entire process was autonomous, all the 116,000 software lines, as these were called, were reviewed and improvements made in addition to widening the dispersion and deviations that it could handle. Somanath even had a separate team from VSSC that was not involved with the software systems to independently check all parameters and ensure that all the bugs were fixed. Then to see if the guidance control systems worked, Veeramuthuvel had the team hire a helicopter and fly a model of the Vikram. It hovered over the ground at a height of 150 metres, the same at which it would on the moon. To avoid the control issues of